Portfolio

(for more information, please visit https://nichoho.github.io/portfolio/)

I. Skills

1. Frontend Development

Tech Stack: React.js, Next.js, Tailwind, CSS, Bootstrap, jQuery, HTML5, CSS3
Specializing in building modern, responsive web interfaces and high-performance landing pages. Focus on creating seamless user experiences using the latest frontend frameworks.

2. Backend Development

Tech Stack: ASP.NET, Laravel, Flask, Node.js, C#, PHP, SQL Server, REST APIs
Specializing in robust API architectures using C#. Experienced in architecting complex Warehouse Management Systems (WMS), inventory, and tracking systems. Have also built scalable API services using Flask and Laravel.

3. Artificial Intelligence

Tech Stack: TensorFlow, Keras, OpenCV, Scikit-learn, LangChain, FAISS, HuggingFace, SHAP
Engineering intelligent systems including RAG pipelines for document retrieval, real-time Computer Vision for sign language recognition, and predictive modeling for sports analytics. Published researcher in Explainable AI, applying predictive modeling to healthcare diagnostics with a focus on model interpretability.

II. Work Experience

1. Lead Developer

Oct 2025 - Present
Nexus Software Agency

- Spearheaded the development of high-performance landing pages for diverse clients using Next.js
- Translated business requirements into modern, scalable frontend code

Tech Stack: Next.js, Tailwind, CSS, React

2. Full Stack Developer

Apr 2024 - Present
Galva Group

- Architected a comprehensive Warehouse Management System (WMS) using ASP.NET.
- Developed a comprehensive Transport Tracker system using React for the frontend and C# Web API for the backend to monitor product logistics in real-time.
- Optimized database queries and frontend performance using jQuery.

Tech Stack: C#, ASP.NET, React, HTML, CSS, SQL Server, JavaScript, jQuery

3. **Full Stack Developer**
   May 2023 - Aug 2023
   Galva Group
   - Developed responsive internal sales dashboards using ASP.NET & Bootstrap
   - Architected scalable Backend services using C# and ASP.NET Core.
   - Designed complex backend logic to handle sales data processing.

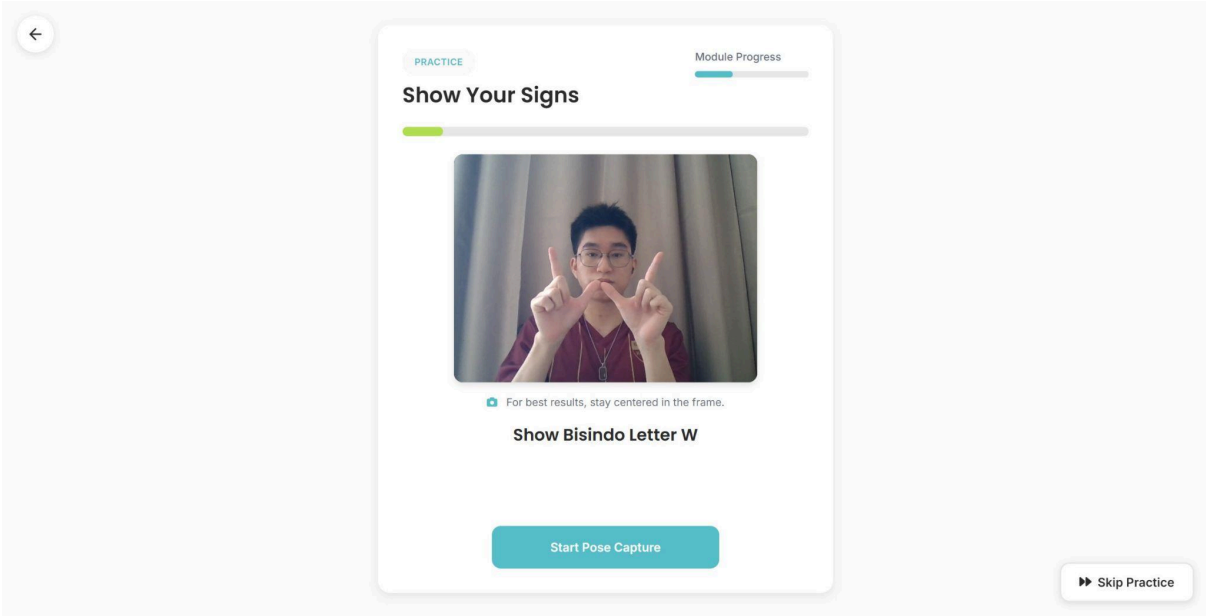
   Tech Stack: C#, ASP.NET, HTML, CSS, Bootstrap, SQL Server, JavaScript, jQuery

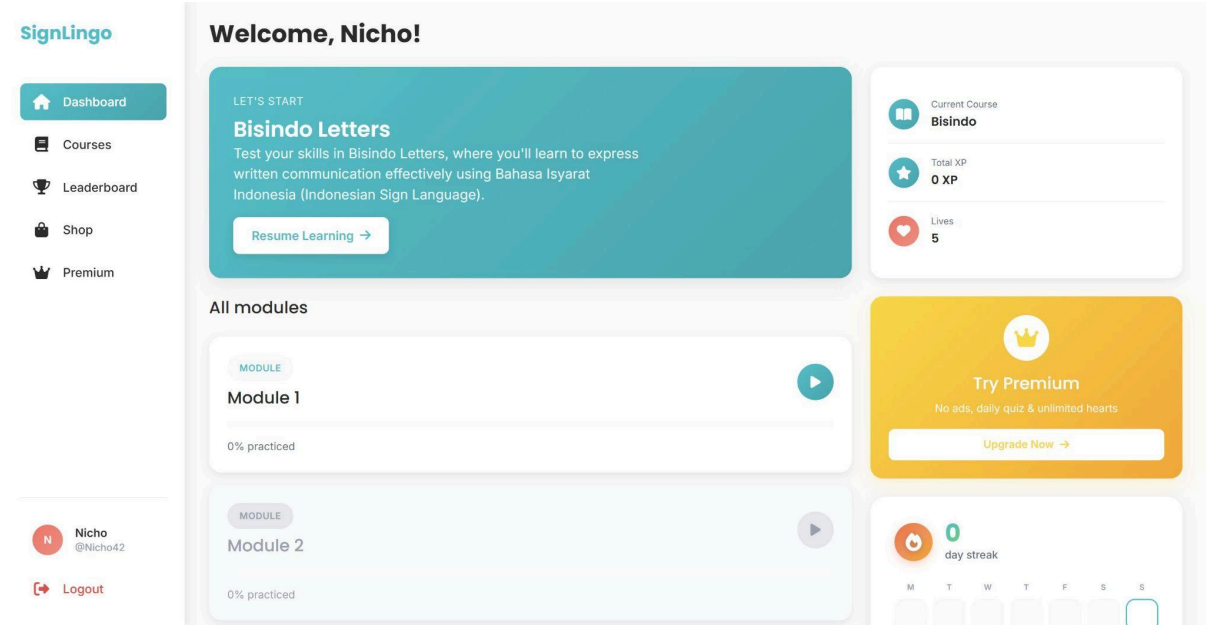
4. **Software Developer**
   June 2022 - July 2022
   Galva Group
   - Engineered a location-tracking CRUD application integrating Google Maps API
   - Built real-time chat features using VB.NET.

   Tech Stack: VB.NET, SQL Server, Google Maps API

III. Projects

1. **Signlingo**





Live Link: https://signlingo-django.onrender.com
Github Link: https://github.com/NichoHo/Signlingo
Signlingo utilizes multimedia components including video tutorials, image-based quizzes, and a real-time AI-powered hand sign recognition game. The system analyzes user signs via webcam to provide instant feedback, bridging the gap between theoretical knowledge and practical application.

2. Flux Budget App

flux

- Dashboard
- Transactions
- Recurring Bills
- Budget
- Analytics
- Settings

Logout

Welcome back, Nicholas Ho

Track your income and expenses

USD ($) | + Add Transaction

Total Balance
$ -172.92

Total Income
$ 0.00

Total Expenses
$ 172.92

Recent Activity

Date	Description	Category	Amount	Receipt
2025-12-23	Dokter	Healthcare	$ 157.28	-
2025-12-22	Netflix (Recurring)	Entertainment	$ 5.23	-
2025-12-16	Makan	Food	$ 10.41	

View Full History →

flux

- Dashboard
- Transactions
- Recurring Bills
- Budget
- Analytics
- Settings

Logout

Transactions

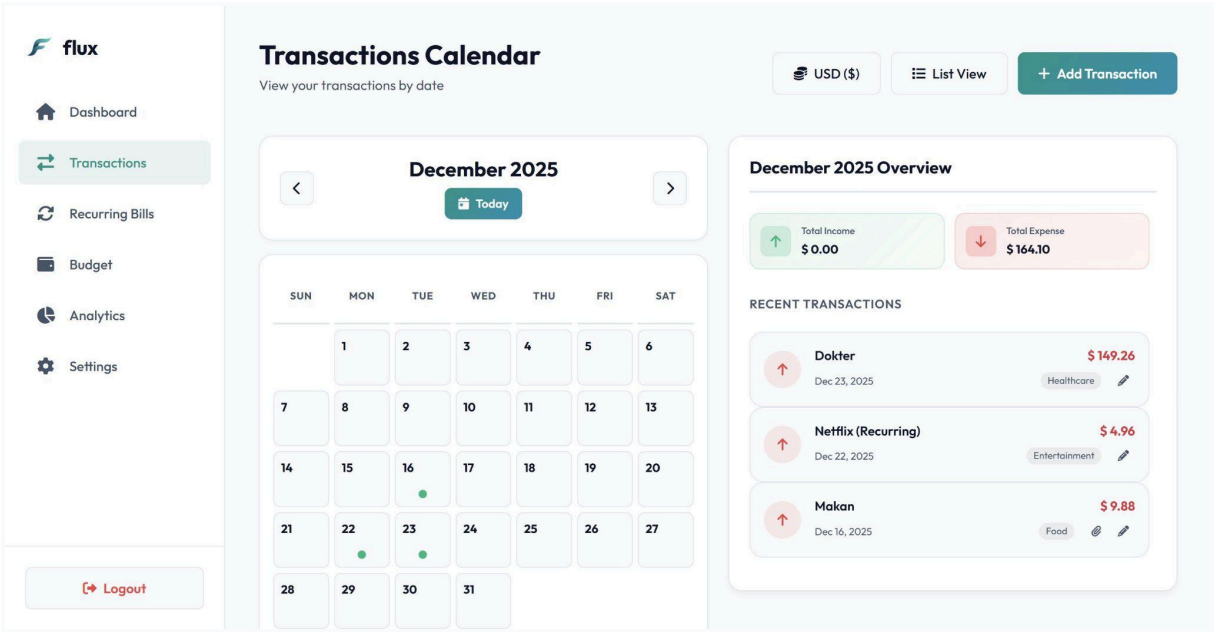
Track your income and expenses

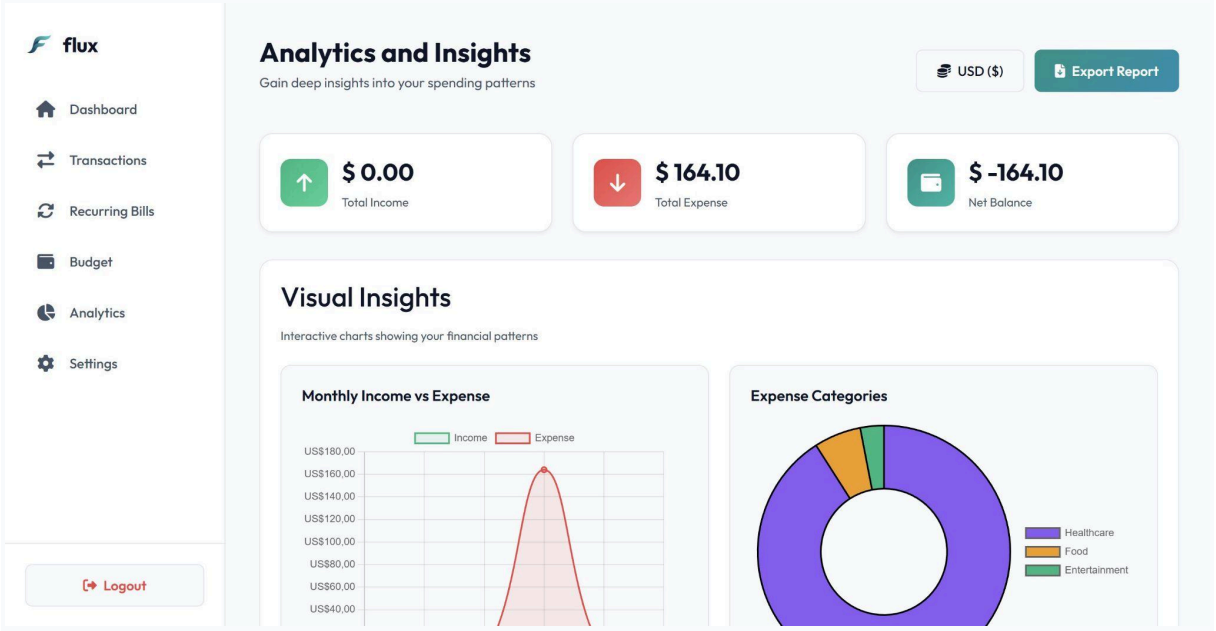
USD ($) | Transactions Calendar | + Add Transaction

Description: e.g. Starbucks, Rent...
Date From: mm/dd/yyyy
Date To: mm/dd/yyyy
More | Apply

Transactions: 3 | Total Volume: $ 172.92

DATE	DESCRIPTION	CATEGORY	RECEIPT	AMOUNT
2025-12-23	**Dokter**	Healthcare	-	$ 157.28
2025-12-22	**Netflix (Recurring)**	Entertainment	-	$ 5.23
2025-12-16	**Makan**	Food	View	$ 10.41

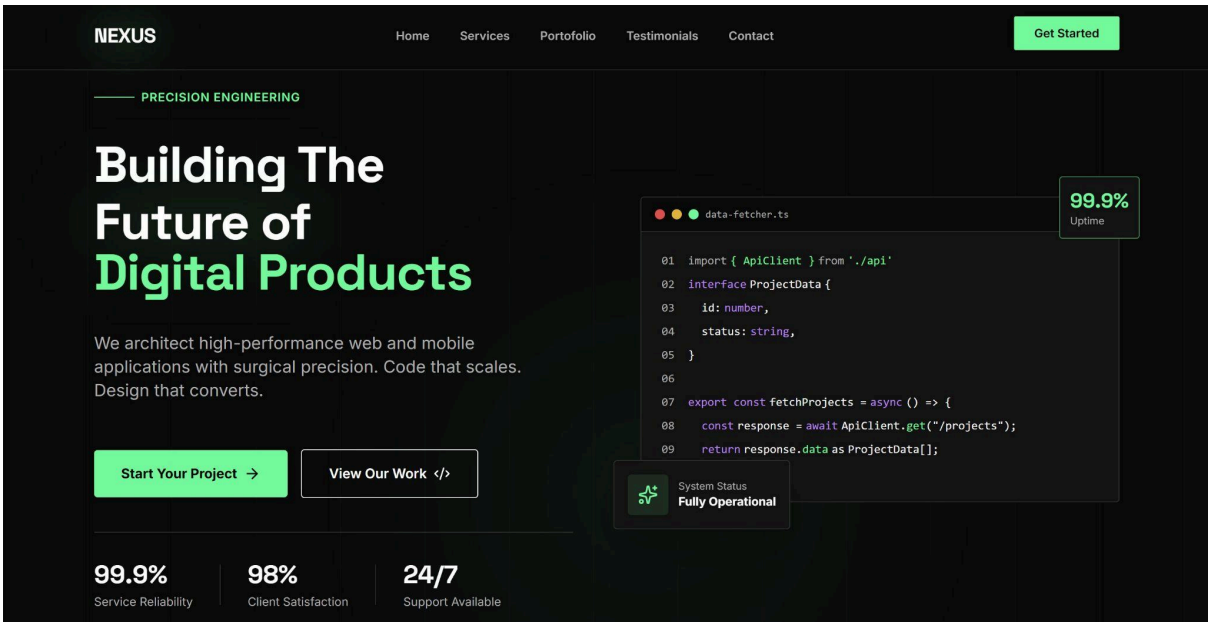


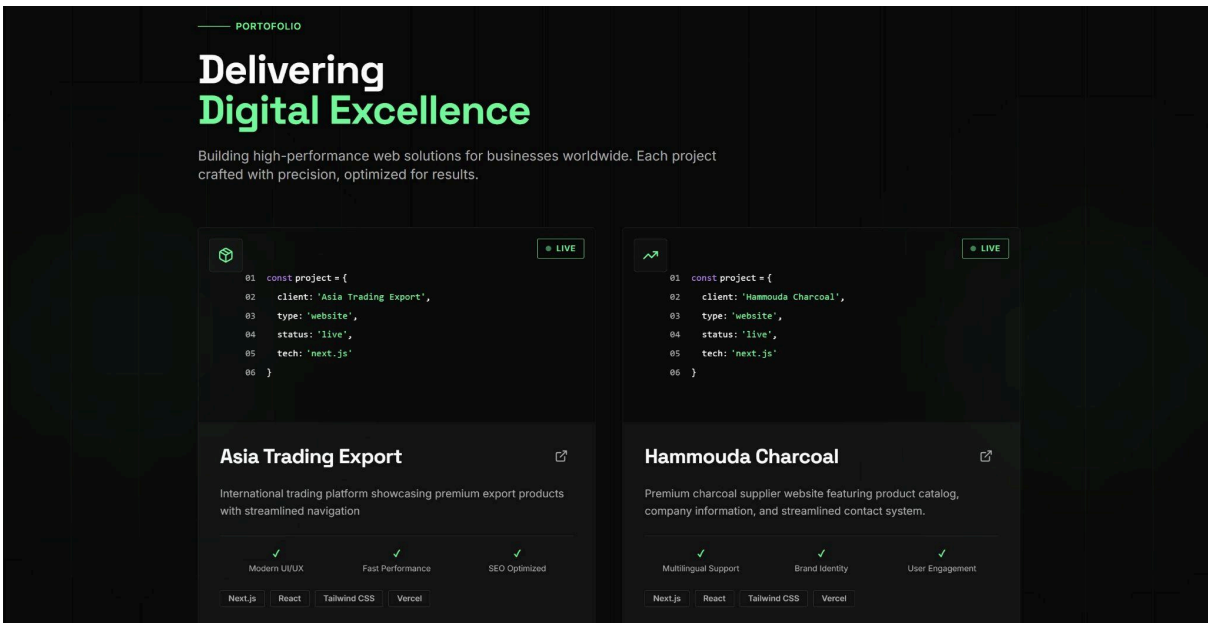


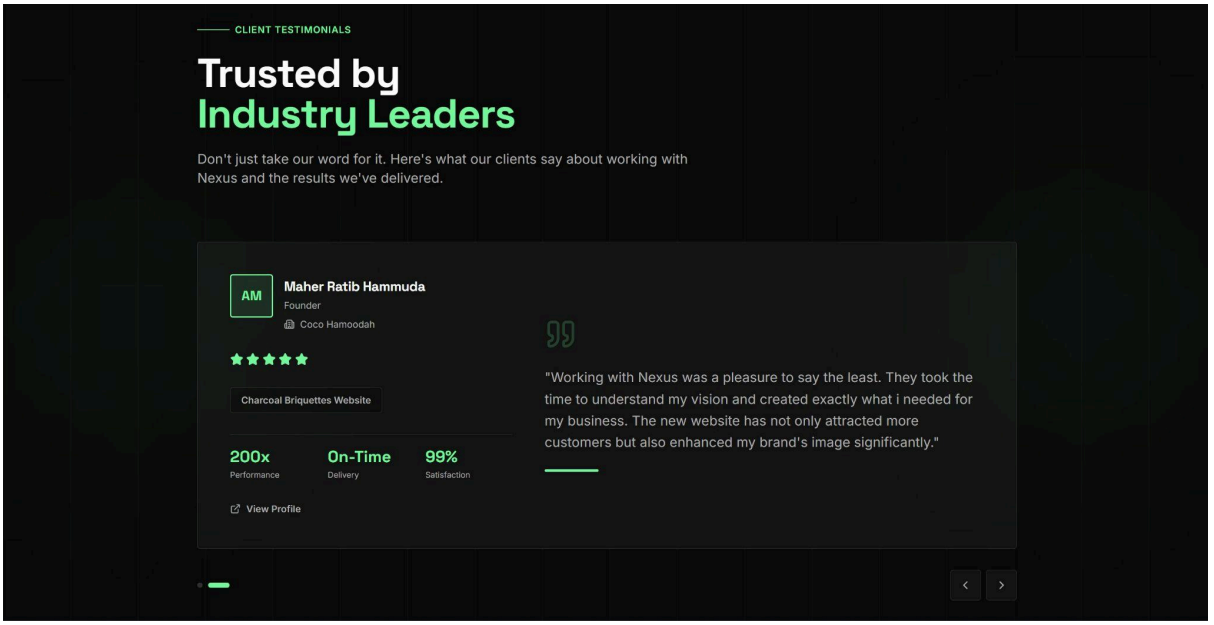
Github Link: https://github.com/NichoHo/Flux_Budget_App

Flux is designed to solve the complexity of managing personal finances across multiple currencies and categories. It moves beyond simple expense tracking by incorporating automated background jobs for recurring bills, budget limit enforcement, and dynamic currency conversion.

3. Nexus Agency

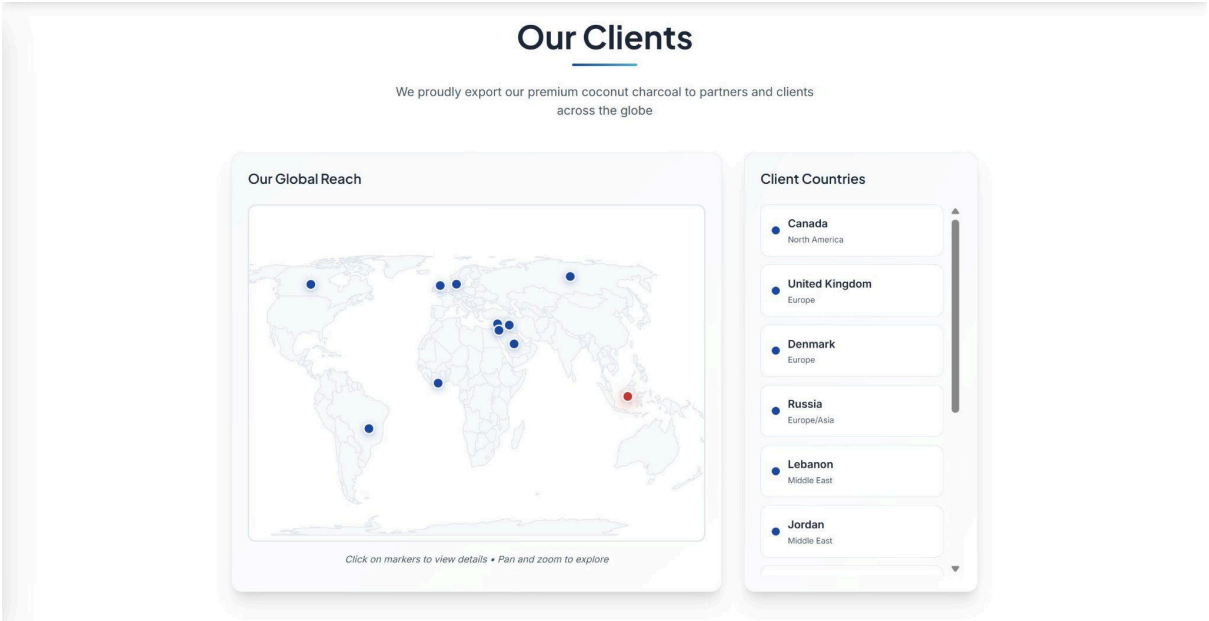






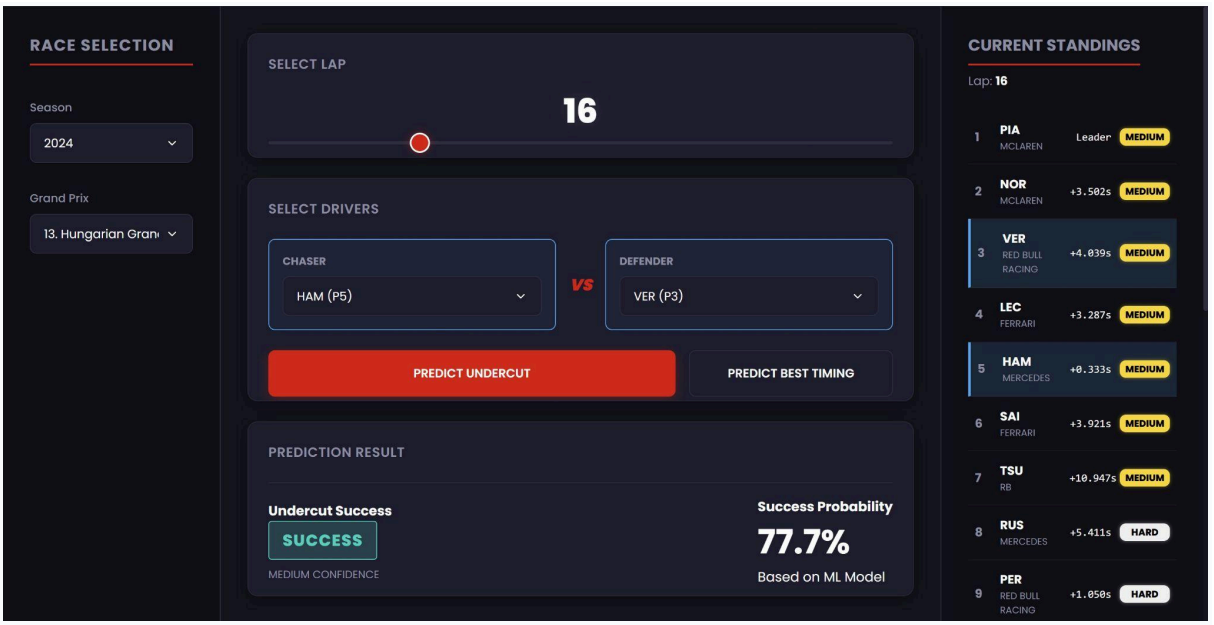
Live Link: https://www.nexdevsoftware.com/
Github Link: https://github.com/NichoHo/nexus-agency
Nexus serves as a comprehensive portfolio and service platform for a digital agency. The project focuses on "Technical Aesthetics", using clean code to drive a visually stunning user experience. It leverages the latest Next.js 14 App Router for performance and Framer Motion for sophisticated, scroll-triggered animations that guide the user journey.

4. Asia Trading Export



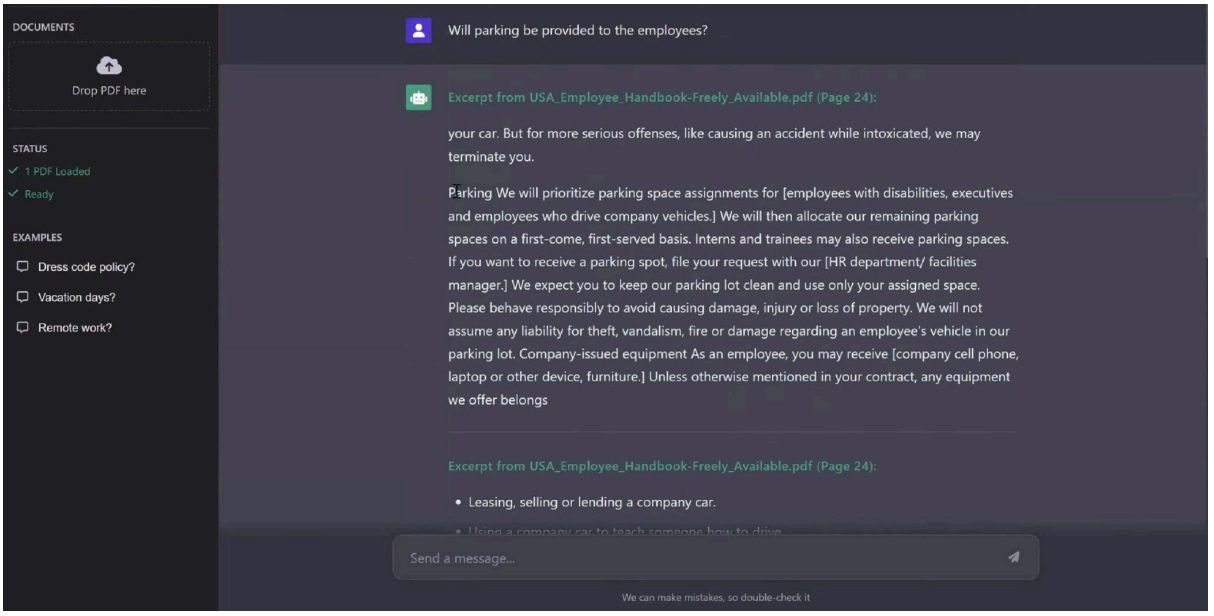
Live Link: https://www.asiatradingexport.com/
Github Link: https://github.com/NichoHo/asia-trading-export
A premium B2B digital platform connecting Indonesian charcoal producers with global markets. Features interactive data visualizations using D3.js.

5. F1 Undercut Predictor



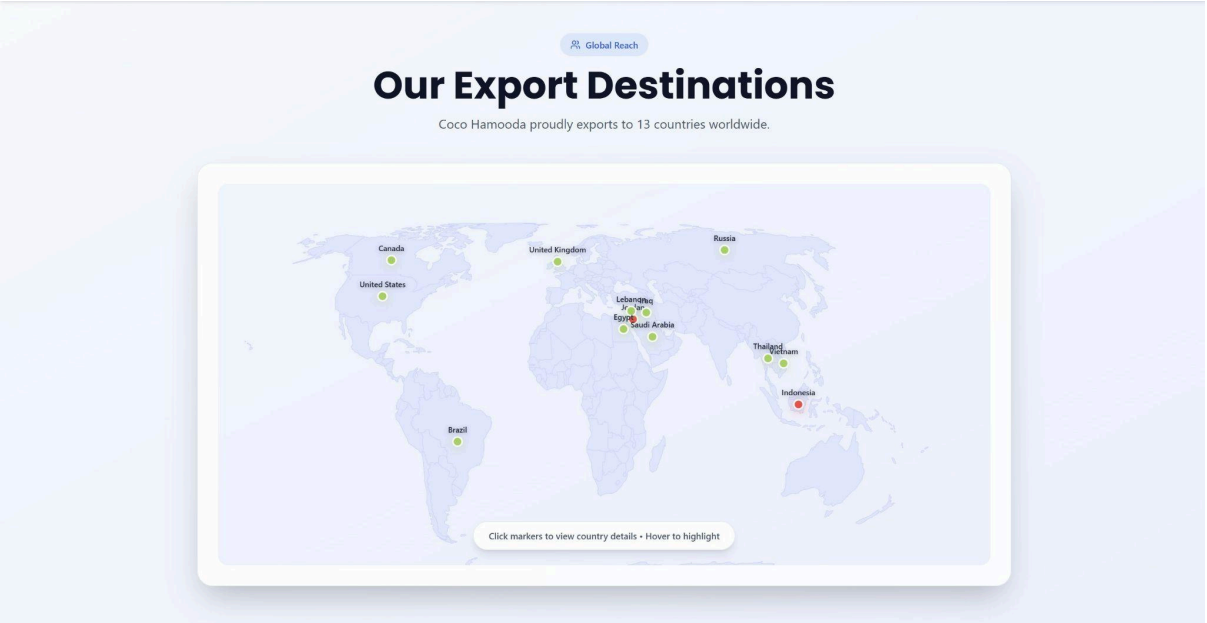
Github Link: https://github.com/NichoHo/f1-undercut-predictor
In Formula 1, an "undercut" is a strategic pit stop made to overtake a rival by utilizing fresh tires. This project leverages the FastF1 library to mine thousands of laps of historical race data (2022-2024). It uses a Random Forest Classifier to calculate the probability of a successful overtake based on variables like tire degradation, pit stop time loss, and gap to the driver ahead.

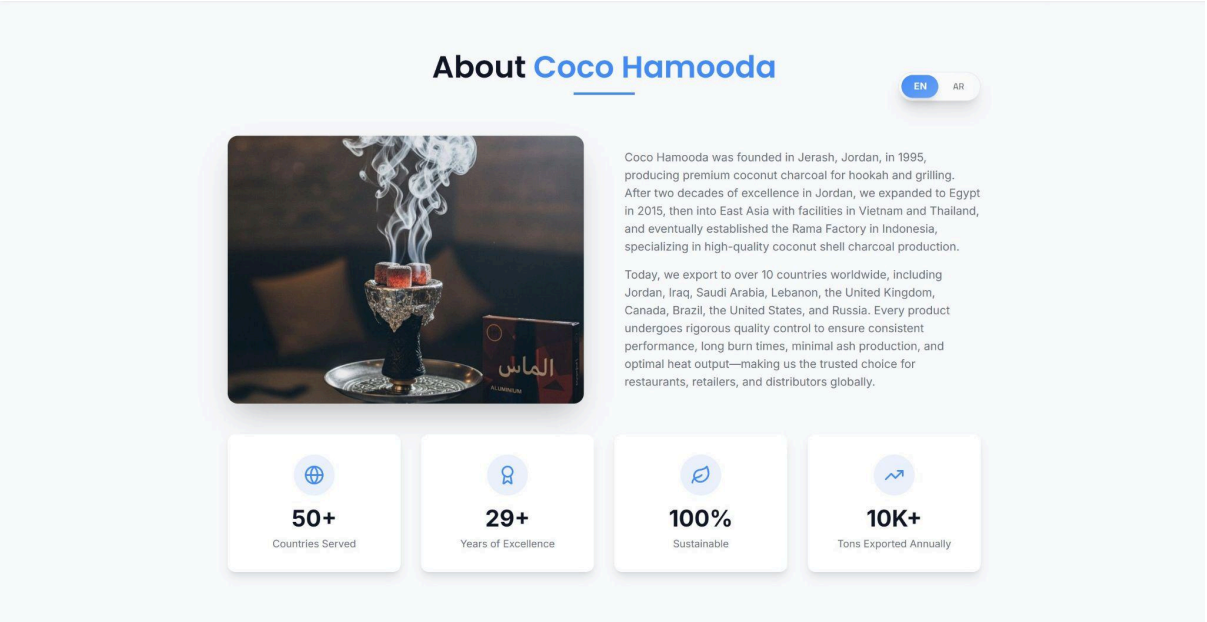
6. FaQ Assistant



Github Link: https://github.com/NichoHo/faq-assistant
FaQ Assistant bridges the gap between static documents and dynamic information retrieval. By uploading a PDF, the system creates semantic embeddings of the content, allowing users to ask natural language questions. The app retrieves the most relevant context and uses a Large Language Model (Gemini 1.5 Flash) to generate accurate, context-aware answers.

7. Hammouda Charcoal





Live Link: https://www.hammoudacharcoal.com/
Github Link: https://github.com/NichoHo/coco-hamooda
This project serves as the primary digital touchpoint for Hammouda Charcoal, facilitating B2B connections between Indonesian manufacturers and Middle Eastern/Global markets. It moves beyond a simple landing page, offering a multi-page structure that details the company's manufacturing process, client portfolio, and extensive product specifications.

IV. Research & Publication

Machine Learning Algorithms for HIV/AIDS Prediction using Explainable AI
ICORIS 2025
Conducted in-depth research on applying Explainable Artificial Intelligence (XAI) to medical diagnostics. The study evaluates various ML models to predict HIV/AIDS susceptibility based on socio-behavioral data, using SHAP (SHapley Additive exPlanations) to provide

transparent, interpretable reasoning for model decisions. This work aims to bridge the trust gap between AI systems and medical practitioners.
Tech Stack: Explainable AI (XAI), Healthcare Informatics, SHAP Analysis, Python, Machine Learning

V. Education

1. **Sejong University**
   Feb 2026 - Aug 2026
   Currently participating in a student exchange program, expanding global perspectives and adapting to an international academic environment.

2. **Binus University**
   2023 - Present
   GPA: 3.83
   - Freshmen Leader & Partner
     Sep 2024 - Present · 1 yr 6 mos
     Guided freshmen through their first year, delivering presentations and providing ongoing mentorship to help them adapt to university life and academic expectations.
   - Staff of HIMTI KIT Division (TECHNO 2024)
     Jul 2024 - Sep 2024 · 3 mos
     Github Link: https://github.com/NichoHo/Techno2024
     Created the HIMTI KIT (university material notes) and developed the official TECHNO 2024 website, ensuring intuitive navigation and reliable performance.
   - Web Development Division (TECHFEST 2024)
     Apr 2024 - Jul 2024 · 4 mos
     Github Link: https://github.com/NichoHo/techfest2024
     Developed and managed the official TECHFEST 2024 website, playing a key role in delivering a reliable digital platform for participant engagement.

3. **Santa Ursula BSD Highschool**
   2021 - 2023
   Student Council (2022-2023)
   Managed the Student Council's social media accounts, creating content to increase engagement and inform the student body of upcoming events.

VI. Certifications

1. **Alibaba Cloud Associate** (Cloud Engineeer)
2. **Conversational AI** (NVIDIA Deep Learning Institute)
3. **Getting Started with Compute** (AWS Educate)
4. **Introduction to Cloud 101** (AWS Educate)